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Aryabhatta Knowledge University, Patna

B.Tech 6th Semester Examination, 2018

Time: 03 Hours

Code: 051614 ***Subject: Software Engineering*** ***Full Marks: 70***

Instructions:-

(i) All questions carry equal marks.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer any seven of the following as directed: **[14]**

(a) What is throwaway prototype?

(b) What is the most popular model for student program?

(i) Waterfall model
(ii) Built-and-fix model
(iii) Spiral model
(iv) Rational unified model

(c) Which model is suitable for stable and known requirements?

(i) Spiral model
(ii) Prototyping model
(iii) Waterfall model
(iv) Iterative enhancement model

(d) Write any four differences between flowchart and DFD.

(e) Write the name of different categories of S/W maintenance models.

(f) Which model is also known as 'design a little, build a little, test a little, deploy a little' model?

(g) All verification activities are ... testing.

(i) dynamic testing
(ii) functional testing
(iii) structural testing

(h) Boundary value analysis generates x test cases for n variables. The value of x is

(i) $8n+1$
(ii) $4n+1$
(iii) $6n+1$
(iv) $2n+1$

(i) Risk is defined as

(i) probability x priority
(ii) probability x size
(iii) priority x size
(iv) probability x impact

(j) A fault is nothing but a/an

(i) error
(ii) defect
(iii) mistake
(iv) All of the above

Q.2 Solve all questions : **[14]**

(a) What is the principal aim of the software engineering discipline? What does the discipline of software engineering discuss?

(b) Do you agree with the statement, "The emphasis of exploratory programming is on error correction while the software engineering practices emphasize error prevention"? Give reasons to justify your answer.

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Q.3 Solve this question : **[14]**

Which life cycle model you will follow for developing software for each of the following applications? Mention the reasons justifying your choice of a particular life cycle model:

(a) A well-understood data processing application.

(b) A new software product that would connect computers through satellite communication. Assume that your team has no previous experience in developing satellite communication software.

(c) A software product that would function as the controller of a telephone switching system.

(d) New library automation software that would link various libraries in the city.

(e) Extremely large software that would provide, monitor and control cellular communication among its subscribers using a set of revolving satellites.

(f) A new text editor.

(g) A compiler for a new language.

Q.4 Solve both questions : **[14]**

(a) Discuss the significance and use of requirement engineering. What are the problems in the formulation of requirements?

(b) Explain the importance of requirements. How many types of requirements are possible and why?

Q.5 Solve all questions : **[14]**

(a) What is project management? What are the characteristics that define a project? Differentiate between project and process management.

(b) Is it true that a software product can always be developed faster by having a larger development team of competent software engineers? Justify your answer.

Q.6 Solve this question : **[14]**

Suppose you are developing a software product in the organic mode. You have estimated the size of the product to be about 100000 lines of code. Compute the nominal effort and the development time.

Q.7 Solve this question : **[14]**

What is software testing? Design the black-box test suite for a function named quadratic solver. Quadratic solver accepts the three floating-point numbers (a, b, c) represent a quadratic equation of the form $ax^2+bx+c=0$. It computes and displays the solution.

Q.8 Write short notes on any two of the following: **[14]**

(a) Software reuse

(b) CASE

(c) Software maintenance

Q.9 Solve both questions : **[14]**

(a) What is software quality? Discuss software quality attributes.

(b) Discuss the differences between object-oriented and function-oriented designs.

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